



Intellectual property (IP) data analytics for innovation, research and development, and strategic management – Adopting state-of-the-art artificial intelligent (AI) and machine learning (ML) approaches

Editor's point

Integrating artificial intelligence (AI) and large language models (LLMs) into the intellectual property landscape had significant impacts in recent years. Add 1-2 Examples and References here. Besides practical impact, substantial scholarly attention has also been devoted to the topic [1].

In this special issue, we focus on these innovative solutions and impacts of artificial intelligence and machine learning in the domains of patent data analytics and IP management from different perspectives. This includes not only the novel methodologies and applications of AI in patent data retrieval, processing, analysis, and management, but also depicting the impacts of AI-related IP policies, related IP laws, regulations, and legal applicability. Investigating new AI techniques and applications to assist inventors and IP applicants in smart application processes is among the key topics in the resulting special issue. It seeks to provide comprehensive consideration, also critically addressing the impact of artificial intelligence in intellectual property ecosystems. Submitted articles should primarily focus on the AI techniques, solutions, applications, and impacts on IP data analytics and patent informatics for innovation, research and development, and strategic management.

With respect to patent and trademark similarity analysis, the maturation of large language models (LLMs) has encouraged firms to explore their use in automated patent comparison to support competitive intelligence and research and development strategy formulation [2]. However, prior studies have noted that LLM-generated judgments may still suffer from bias, inconsistency, and limited decision transparency. To address these challenges, Choi and Park [3] propose a three-dimensional comparative judgment framework, in which prompt refinement is employed to enhance the reliability and interpretability of LLM-based patent comparison.

In parallel, building upon existing evidence supporting the effectiveness of (Subject-Action-Object (SAO) semantic structures in patent similarity analysis [4], Wang et al. [5] further develop optimized semantic analysis methods to improve data usability and analytical performance. In addition, Reis et al. [6] apply LLMs to trademark similarity assessment, demonstrating the extended potential of generative AI in intellectual property analysis, particularly for non-patent textual data.

Regarding sustainable development and industrial transformation, amid the accelerating global push toward carbon neutrality and digital transformation, sustainable technologies under Industry 5.0 are progressively reshaping future manufacturing paradigms. Nevertheless,

existing research has largely focused on conceptual or macro-level discussions, such as definitions, characteristics, drivers, challenges, and opportunities [7,8]. Addressing this gap, Yang and Chen [9] conduct a patent-based analysis to examine the life-cycle evolution of Industry 5.0-related technologies, providing empirical insights to inform policy formulation and industrial strategy. Complementarily, effective management of intellectual property is essential for organizations in competitive sectors like FinTech [10]. Armani Dehghani et al. [11] focus on this highly competitive FinTech sector, introducing an AI-driven patent analytics approach as a forward-looking tool to support innovation management and strategic intellectual property planning.

In terms of foundational patent analytics tasks, including patent retrieval, summarization, and classification, the rapid growth in global patent filings has made the effective handling of large-scale patent data increasingly challenging. Although the potential of generative AI for tasks such as legal document summarization has attracted growing attention, in-depth studies remain relatively limited [12]. In response, Trappey et al. [13] develop an intelligent patent summarization system leveraging LLM technology to enhance the comprehensibility and practical usability of patent documents.

Moreover, patent classification continues to face challenges arising from data scale, technical complexity, and overlapping technological domains. The accuracy and timeliness of patent classification extend beyond administrative efficiency by directly influencing competitive intelligence, innovation policy, and strategic resource allocation [14]. To address these issues, Xiong et al. [15] propose a multi-label patent classification framework that integrates the text understanding capabilities of LLMs with active learning, thereby mitigating the limitations of existing manual and automated systems in terms of scalability, cost, and classification accuracy.

With respect to generative AI and advanced patent knowledge applications, ongoing advances in AI technologies have increased attention to AI-assisted invention and knowledge generation. Lee's study [16] demonstrates the feasibility of using AI to generate patent claims exhibiting higher levels of novelty. Meanwhile, Lin and Chou [17] integrate bibliometric analysis, BERT-based topic modeling, and systematic literature review to construct a comprehensive matrix aligning distinct AI techniques with corresponding patent analysis tasks, thereby systematically mapping the practical application landscape of AI in patent analytics. Additionally, Siddharth [18] introduces a novel methodology based on patent knowledge graphs to measure and tune the modularity of technological artifacts, opening new avenues for patent-driven engineering design research.

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Overall, the studies collected in this special issue reveal that AI is evolving from an auxiliary analytical tool into a structurally embedded and strategically oriented core component of intellectual property analysis. The contributions extend beyond foundational tasks such as summarization, classification, and similarity detection, encompassing technology life-cycle analysis, innovation strategy support, and generative knowledge creation. Collectively, this special issue highlights the emergence of an integrated research trajectory that bridges technological and managerial perspectives, providing a robust foundation for future research and practice in AI-driven intellectual property management and innovation decision-making and its effects on innovation, research and development, and strategic management.

CRedit authorship contribution statement

Chin-Yuan Fan: Writing – review & editing. **Chun-Yi Wu:** Writing – review & editing. **Carsten C. Guderian:** Writing – review & editing. **Shu-Hao Chang:** Writing – review & editing, Writing – original draft. **Anthony Trippe:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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